

Model1::ClassDiagram1

```
classDiagram
    class HumanBody {
        <<abstract>>
        -Age: Integer
        -BloodType: String
        -OrgansNumber: Integer
        -Height: Integer
        -Weight: Integer
        -BMI: Integer
        +Sleep(): Boolean
        +ResponseToStimulus(): String
    }
    class Sex {
        <<enumeration>>
        Male
        Female
    }
    class LivingBeings {
        -Species: String
    }
    class IntegumentarySystem {
        -Hair: Integer
        -Nails: Integer
        +protection(): Boolean
    }
    class MusculoskeletalSystem {
        -Bones: Integer
        -SkeletalMuscles: Integer
        +Move(): Boolean
    }
    class RespiratorySystem {
        -Lungs: Integer
        -RespirationMuscles: Integer
        -Inhalation(): Boolean
        -Exhalation(): Boolean
    }
    class CirculatorySystem {
        -Heart: String
        -BloodVessels: Integer
        +FluidCirculation(): Boolean
    }
    class DigestiveSystem {
        -Stomach: Integer
        -Intestines: Integer
        +FoodBreakdown(): Boolean
    }
    class ExcretorySystem {
        -Kidneys: Integer
        -Ureters: Integer
        +SubstancesRemoval(): Boolean
    }
    class NervousSystem {
        -Nerves: Integer
        -SpinalCord: Integer
        +MuscularResponses(): Boolean
    }
    class EndocrineSystem {
        -Hormones: String
        +RegulateMetabolism(): Boolean
    }
    class ReproductiveSystem {
        -FertilityLevel: Integer
        +Reproduction(): Boolean
    }
    class Skin {
        <<abstract>>
        -Color: String
        -Density: Integer
        +ProduceOil(): Integer
    }
    class Oily
    class Dry
    class Normal
    class Air {
        -Oxygen: Integer
        -Hydrogen: Integer
        -Nitrogen: Integer
    }
    class Blood {
        -Type: Integer
        -CoagulationLevel: Integer
        +TransportNutrients(): Boolean
    }

    HumanBody <|-- Sex
    HumanBody <|-- LivingBeings
    HumanBody <|-- IntegumentarySystem
    HumanBody <|-- MusculoskeletalSystem
    HumanBody <|-- RespiratorySystem
    HumanBody <|-- CirculatorySystem
    HumanBody <|-- DigestiveSystem
    HumanBody <|-- ExcretorySystem
    HumanBody <|-- NervousSystem
    HumanBody <|-- EndocrineSystem
    HumanBody <|-- ReproductiveSystem
    HumanBody <|-- Skin
    Skin <|-- Oily
    Skin <|-- Dry
    Skin <|-- Normal
    HumanBody ..> Air
    HumanBody ..> Blood
```

The diagram illustrates the structure of a human body model using UML class notation. The central class is **Human Body**, an abstract class with attributes: Age (Integer), BloodType (String), OrgansNumber (Integer), Height (Integer), Weight (Integer), BMI (Integer), and methods: Sleep() (Boolean), ResponseToStimulus() (String). A note specifies the BMI calculation: $BMI = Weight / (Height)^2$. **Human Body** has several subclasses: **Sex** (an enumeration with Male and Female), **Living Beings** (with Species: String), and various organ systems: **Integumentary System**, **Musculoskeletal System**, **Respiratory System**, **Circulatory System**, **Digestive System**, **Excretory System**, **Nervous System**, **Endocrine System**, and **Reproductive System**. The **Integumentary System** is further abstracted into **Skin**, which has subclasses **Oily**, **Dry**, and **Normal**. **Human Body** also has dashed associations with **Air** and **Blood**.

